

Deep Learning 2020 Midterm exam_B063012054_林祐安

1. 訓練架構：

三個同架構模型，優化器與 batch size 不同，並行預測，取多數決當預測值。

Layer (type)	Output Shape	Param #	Connected to
input_34 (InputLayer)	(None, 48, 140, 3)	0	
conv2d_326 (Conv2D)	(None, 48, 140, 8)	1952	input_34[0][0]
conv2d_327 (Conv2D)	(None, 48, 140, 8)	5192	conv2d_326[0][0]
batch_normalization_261 (Batch Normalization)	(None, 48, 140, 8)	32	conv2d_327[0][0]
activation_242 (Activation)	(None, 48, 140, 8)	0	batch_normalization_261[0][0]
conv2d_328 (Conv2D)	(None, 48, 140, 16)	10384	activation_242[0][0]
conv2d_329 (Conv2D)	(None, 48, 140, 16)	20752	conv2d_328[0][0]
batch_normalization_262 (Batch Normalization)	(None, 48, 140, 16)	64	conv2d_329[0][0]
activation_243 (Activation)	(None, 48, 140, 16)	0	batch_normalization_262[0][0]
max_pooling2d_91 (MaxPooling2D)	(None, 23, 69, 16)	0	activation_243[0][0]
conv2d_330 (Conv2D)	(None, 23, 69, 32)	41504	max_pooling2d_91[0][0]
conv2d_331 (Conv2D)	(None, 23, 69, 32)	82976	conv2d_330[0][0]
batch_normalization_263 (Batch Normalization)	(None, 23, 69, 32)	128	conv2d_331[0][0]
activation_244 (Activation)	(None, 23, 69, 32)	0	batch_normalization_263[0][0]
conv2d_332 (Conv2D)	(None, 23, 69, 64)	18496	activation_244[0][0]
conv2d_333 (Conv2D)	(None, 23, 69, 64)	36928	conv2d_332[0][0]
batch_normalization_264 (Batch Normalization)	(None, 23, 69, 64)	256	conv2d_333[0][0]
activation_245 (Activation)	(None, 23, 69, 64)	0	batch_normalization_264[0][0]
dropout_67 (Dropout)	(None, 23, 69, 64)	0	activation_245[0][0]
max_pooling2d_92 (MaxPooling2D)	(None, 11, 34, 64)	0	dropout_67[0][0]
conv2d_334 (Conv2D)	(None, 11, 34, 128)	73856	max_pooling2d_92[0][0]
conv2d_335 (Conv2D)	(None, 11, 34, 128)	147584	conv2d_334[0][0]
batch_normalization_265 (Batch Normalization)	(None, 11, 34, 128)	512	conv2d_335[0][0]
activation_246 (Activation)	(None, 11, 34, 128)	0	batch_normalization_265[0][0]
conv2d_336 (Conv2D)	(None, 11, 34, 256)	295168	activation_246[0][0]
conv2d_337 (Conv2D)	(None, 11, 34, 256)	590080	conv2d_336[0][0]
batch_normalization_266 (Batch Normalization)	(None, 11, 34, 256)	1024	conv2d_337[0][0]
activation_247 (Activation)	(None, 11, 34, 256)	0	batch_normalization_266[0][0]
dropout_68 (Dropout)	(None, 11, 34, 256)	0	activation_247[0][0]
max_pooling2d_93 (MaxPooling2D)	(None, 5, 16, 256)	0	dropout_68[0][0]
average_pooling2d_33 (AveragePooling2D)	(None, 2, 8, 256)	0	max_pooling2d_93[0][0]
flatten_34 (Flatten)	(None, 4096)	0	average_pooling2d_33[0][0]
digit1 (Dense)	(None, 19)	77843	flatten_34[0][0]
digit2 (Dense)	(None, 19)	77843	flatten_34[0][0]
digit3 (Dense)	(None, 19)	77843	flatten_34[0][0]
digit4 (Dense)	(None, 19)	77843	flatten_34[0][0]
Total params: 1,638,260			
Trainable params: 1,637,252			
Non-trainable params: 1,008			

2. 預處理方法：

使用 CV2 去除雜訊，計算回歸去除線條，並用多進程加速，原本 5000 張圖需處理 11 小時，縮短為 712 秒

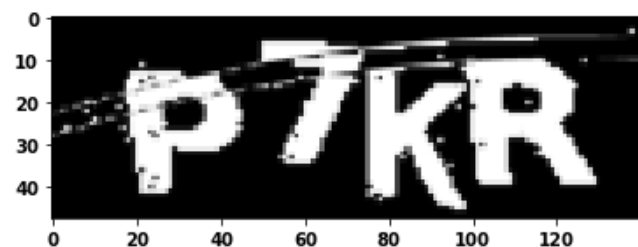
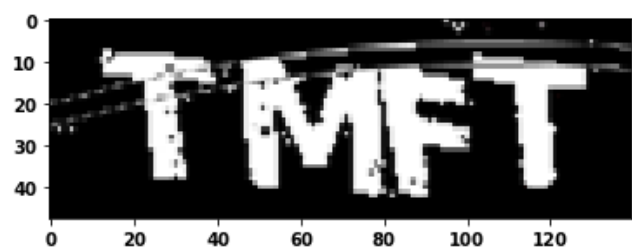
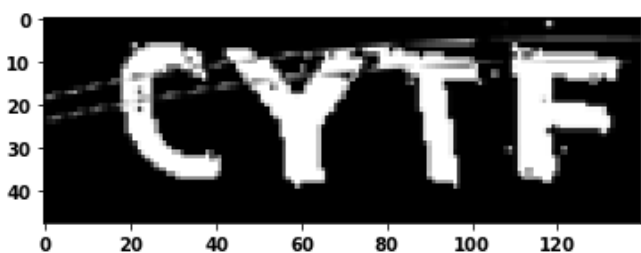
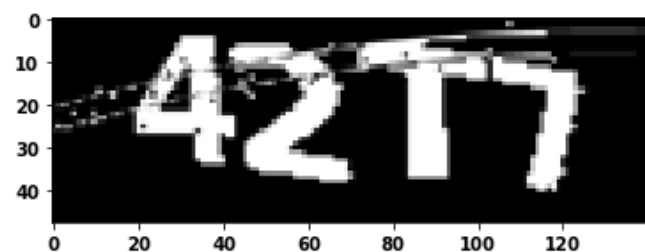
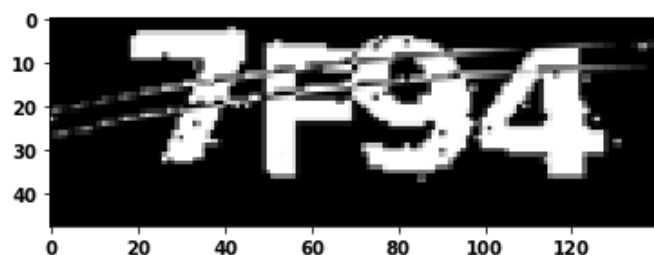
3. 訓練正確率：

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Epoch 1/1
3750/3750 [=====] - 18s 5ms/step - loss: 0.4522
digit1_loss: 0.0137 - digit2_loss: 0.0087 - digit3_loss: 0.0095 - digit4_loss: 0.0058
digit1_accuracy: 0.9973 - digit2_accuracy: 0.9987 - digit3_accuracy: 0.9979 - digit4_accuracy: 0.9997
val_loss: 0.6770 - val_digit1_loss: 0.0877 - val_digit2_loss: 0.0759 - val_digit3_loss: 0.0608 - val_digit4_loss: 0.0580
val_digit1_accuracy: 0.9928 - val_digit2_accuracy: 0.9944 - val_digit3_accuracy: 0.9984 - val_digit4_accuracy: 0.9992
Epoch 1/1
3750/3750 [=====] - 27s 7ms/step - loss: 0.3312
digit1_loss: 0.0097 - digit2_loss: 0.0080 - digit3_loss: 0.0082 - digit4_loss: 0.0062
digit1_accuracy: 0.9981 - digit2_accuracy: 0.9997 - digit3_accuracy: 0.9989 - digit4_accuracy: 1.0000
val_loss: 0.7784 - val_digit1_loss: 0.1320 - val_digit2_loss: 0.1377 - val_digit3_loss: 0.1288 - val_digit4_loss: 0.0996
val_digit1_accuracy: 0.9904 - val_digit2_accuracy: 0.9928 - val_digit3_accuracy: 0.9960 - val_digit4_accuracy: 0.9984
Epoch 1/1
3750/3750 [=====] - 13s 3ms/step - loss: 0.1728
digit1_loss: 0.0255 - digit2_loss: 0.0196 - digit3_loss: 0.0158 - digit4_loss: 0.0120
digit1_accuracy: 0.9941 - digit2_accuracy: 0.9971 - digit3_accuracy: 0.9971 - digit4_accuracy: 0.9984
val_loss: 0.5393 - val_digit1_loss: 0.0921 - val_digit2_loss: 0.1374 - val_digit3_loss: 0.1249 - val_digit4_loss: 0.0825
val_digit1_accuracy: 0.9904 - val_digit2_accuracy: 0.9864 - val_digit3_accuracy: 0.9952 - val_digit4_accuracy: 0.9992
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4. 測試正確率：

3000 picture total wrong=91
 one word accuracy= 0.9924166666666666
 four word accuracy= 0.9756666666666667

5. 測試結果：



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prdict= ['7', 'F', '9', '4']
answer= ['7', 'F', '9', '4'] ,wrong words= 0
prdict= ['4', '2', 'T', '7']
answer= ['4', '2', 'T', '7'] ,wrong words= 0
prdict= ['C', 'Y', 'T', 'F']
answer= ['C', 'Y', 'T', 'F'] ,wrong words= 0
prdict= ['T', 'M', 'F', 'T']
answer= ['T', 'M', 'F', 'T'] ,wrong words= 0
prdict= ['P', '7', 'K', 'R']
answer= ['P', '7', 'K', 'R'] ,wrong words= 0
prdict= ['N', '2', 'Z', 'K']
answer= ['N', '2', 'Z', 'K'] ,wrong words= 0
prdict= ['7', '4', 'F', 'Y']
answer= ['7', '4', 'F', 'Y'] ,wrong words= 0
prdict= ['Z', 'Z', '5', 'C']
answer= ['Z', 'Z', '5', 'C'] ,wrong words= 0
prdict= ['7', 'Z', '4', '4']
answer= ['7', 'Z', '4', '4'] ,wrong words= 0
prdict= ['R', '4', 'Z', 'C']
answer= ['R', '4', 'Z', 'C'] ,wrong words= 0
prdict= ['Y', '2', '5', 'T']
answer= ['Y', '2', '5', 'T'] ,wrong words= 0
prdict= ['F', 'M', '2', '2']
answer= ['F', 'M', '2', '2'] ,wrong words= 0
prdict= ['Y', '4', 'M', '4']
answer= ['Y', '4', 'M', '4'] ,wrong words= 0
prdict= ['M', '2', 'Q', '2']
answer= ['M', '2', 'Q', '2'] ,wrong words= 0
prdict= ['F', 'F', '7', 'N']
answer= ['F', 'F', '7', 'N'] ,wrong words= 0
prdict= ['9', '5', 'R', 'M']
answer= ['9', '5', 'R', 'M'] ,wrong words= 0
prdict= ['T', '4', 'T', '5']
answer= ['T', '4', 'T', '5'] ,wrong words= 0
prdict= ['F', '3', '4', 'A']
answer= ['F', '3', '4', 'A'] ,wrong words= 1
prdict= ['F', 'A', '4', 'A']
answer= ['F', 'A', '4', 'A'] ,wrong words= 0
prdict= ['3', '2', 'H', '4']
answer= ['3', '2', 'H', '4'] ,wrong words= 0
prdict= ['Q', 'H', 'M', 'A']
answer= ['Q', 'H', 'M', 'A'] ,wrong words= 0
prdict= ['P', 'Y', 'C', 'P']
answer= ['P', 'Y', 'C', 'P'] ,wrong words= 0
prdict= ['A', 'P', 'N', 'M']
answer= ['A', 'P', 'N', 'M'] ,wrong words= 0
prdict= ['R', 'M', 'Q', 'N']
answer= ['R', 'M', 'Q', 'N'] ,wrong words= 0
prdict= ['K', 'Z', '7', '4']
answer= ['K', 'Z', '7', '4'] ,wrong words= 0
prdict= ['N', 'F', 'Z', 'H']
answer= ['N', 'F', 'Z', 'H'] ,wrong words= 0
prdict= ['Y', '2', '7', 'R']
answer= ['Y', '2', '7', 'R'] ,wrong words= 0
prdict= ['A', 'Y', 'R', 'K']
answer= ['A', 'Y', 'R', 'K'] ,wrong words= 1
prdict= ['M', '9', '4', 'Y']
answer= ['M', '9', '4', 'Y'] ,wrong words= 0
prdict= ['3', '5', 'Q', '7']
answer= ['3', '5', 'Q', '7'] ,wrong words= 0
prdict= ['A', 'Y', 'H', 'A']
answer= ['A', 'Y', 'H', 'A'] ,wrong words= 0
prdict= ['T', 'R', '4', '5']
answer= ['T', 'R', '4', '5'] ,wrong words= 0
prdict= ['Q', 'T', 'R', '4']
answer= ['Q', 'T', 'R', '4'] ,wrong words= 0
prdict= ['2', '7', '3', 'A']
answer= ['2', '7', '3', 'A'] ,wrong words= 0
prdict= ['Q', 'M', 'Y', 'F']
answer= ['Q', 'M', 'Y', 'F'] ,wrong words= 0
prdict= ['2', 'Y', 'N', 'K']
answer= ['2', 'Y', 'N', 'K'] ,wrong words= 0
prdict= ['R', 'H', 'T', 'P']
answer= ['R', 'H', 'T', 'P'] ,wrong words= 0
prdict= ['7', 'K', 'Q', 'M']
answer= ['7', 'K', 'Q', 'M'] ,wrong words= 1
prdict= ['7', 'H', 'Y', 'P']
answer= ['7', 'H', 'Y', 'P'] ,wrong words= 0
prdict= ['5', 'M', '3', 'K']
answer= ['5', 'M', '3', 'K'] ,wrong words= 0

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